

# Internship Project Report

**Project Title:** AI-Powered Skin Intelligence and Personalized Skincare Routine Recommendation Platform (SkinIQ)

**Role:** AI Engineer Intern

**Date:** October 2026

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## 1. Introduction & Overview

During my internship, I was tasked with developing the **AI Skincare Planner (SkinIQ)**, a comprehensive, full-stack platform designed to bridge the gap between everyday consumers and dermatological professionals. The goal was to build a robust, data-driven application capable of generating personalized skincare routines, tracking daily habits, and providing clinical dashboards for consultants and dermatologists.

## 2. Problem Statement

Many consumers struggle to find skincare routines that adapt dynamically to their changing environment, lifestyle, and specific skin conditions. Simultaneously, skincare consultants and dermatologists lack centralized, real-time data to monitor patient progress. A unified platform was needed to collect user telemetry (sleep, hydration, UV index), assess skin health, and securely present this data to medical professionals.

## 3. Technologies Used

To ensure the application was scalable, secure, and maintainable, I utilized the following technology stack:

- Backend:** Python, FastAPI (for rapid, asynchronous API development)
- Database:** MongoDB (NoSQL schema for flexible, dynamic health profiles)
- Frontend:** Vanilla JavaScript, HTML5, CSS3 with Glassmorphism aesthetics
- Data Visualization:** Chart.js (for analytics and progress tracking)
- Infrastructure & Deployment:** Docker, Docker Compose, Git/GitHub, and Render (Cloud Deployment)

## 4. Project Implementation & Milestones

## Milestone 1: Foundation & API Architecture

- Designed the core MongoDB database schemas for `Users` , `SkinProfiles` , and `Assessments` .
- Developed a secure RESTful API using FastAPI, implementing JWT (JSON Web Token) authentication.
- Established Role-Based Access Control (RBAC) to differentiate between Users, Consultants, Dermatologists, and Administrators.

## Milestone 2: Intelligent Recommendation Engine

- Built the recommendation algorithm that factors in skin type (e.g., dry, oily), medical concerns (e.g., acne, rosacea), and ingredient allergies.
- Implemented an ingredient conflict matrix to alert users to dangerous product combinations (e.g., Retinol + Vitamin C).

## Milestone 3: Telemetry & Progress Tracking

- Developed a daily logging system for users to track hydration, sleep quality, and environmental factors like UV index.
- Created visual progress charts in the frontend UI to help users visualize their "Lifestyle Recovery Score" over time.

## Milestone 4: Executive Dashboards, Edge Cases, & Cloud Deployment

- Engineered dedicated portal views for administrative and clinical roles, incorporating dynamic `Chart.js` visual modules (e.g., "System Activity & Engagement" line charts and "Client Assessment" doughnut charts).
- Wrote automated testing scripts ( `edge_case_tester.py` ) to simulate invalid logins, malformed requests, and strict security boundary testing.
- Containerized the entire application utilizing **Docker**, ensuring identical runtime environments from local development to production.
- Configured Infrastructure-as-Code ( `render.yaml` ) for seamless, automated deployment to cloud services.

## 5. Challenges and Solutions

- **Challenge:** Managing distinct user experiences for 4 different roles (Consumer, Consultant, Dermatologist, Admin).

- **Solution:** Implemented strict RBAC on the FastAPI backend and built a dynamic frontend layout that selectively renders navigation items and dashboard canvases based on the decoded JWT token payload.
- **Challenge:** Ensuring deployment stability and environment consistency.
  - **Solution:** Transitioned the architecture to Docker, isolating the Python backend and MongoDB instances. I also mapped out edge case automation to guarantee stability upon pushing updates.

## 6. Conclusion

Building the AI Skincare Planner was an immensely rewarding challenge that allowed me to apply full-stack development, database architecture, and cloud deployment principles in a real-world scenario. I successfully delivered a production-ready, Dockerized platform equipped with clinical dashboards and intelligent algorithms. This project not only solidified my software engineering skills but also demonstrated my ability to own a product lifecycle from conceptualization to cloud deployment.